Huawei Access-Node WFD-ACC-18 Optical Transport Failure SOP

Node: WFD-ACC-18 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_8047

Description

The occurrence of an Optical Transport Failure on Huawei Access-Node WFD-ACC-18 in the Whitefield POP is attributed to the degradation of SFP/XFP transceivers, fiber cuts, or Rx/Tx power loss due to the inherent limitations of the Optical Transport Network (OTN) architecture. The OTN architecture relies heavily on the integrity of the optical path and signal quality, which can be compromised by factors such as fiber attenuation, optical signal-to-noise ratio (OSNR) degradation, or physical damage to the fiber. In the context of Huawei's Access-Node WFD-ACC-18, the OTN architecture is designed to provide high-speed data transmission over optical fibers using the OTU4 protocol. However, the reliability of this architecture is contingent upon the proper functioning of the SFP/XFP transceivers, which can be prone to degradation over time due to environmental factors, usage, or manufacturing defects. This SOP outlines the procedures for diagnosing and resolving Optical Transport Failures on Huawei Access-Node WFD-ACC-18, with a focus on identifying and mitigating the root causes of such failures.

Pre-requisite Checks

- Verify that the Whitefield POP is accessible and that the necessary remote hands are dispatched to the site.

- Check the status of the redundant links and ensure that they are operational and not contributing to the failure.

- Inspect the fiber connections and SFP/XFP transceivers for signs of damage or physical stress.

Potential Root Causes

Cause	Probability
SFP/XFP transceiver degradation	High
Fiber cuts or physical damage	Medium
Rx/Tx power loss due to fiber attenuation or OSNR degradation	Low

Diagnostic Commands & Expected Outputs

```
Command: display opt trans info, Expected Output: OTN transmission information, including optical signal quality and SFP/XFP status, Description: This command provides a comprehensive view of the OTN transmission information, including optical signal quality, SFP/XFP status, and transmission parameters. It helps to identify potential issues related to SFP/XFP degradation or fiber cuts.
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```
Command: display opt trans error-stat, Expected Output: Error statistics for the OTN transmission, including error counts and alarm information, Description: This command provides detailed error statistics for the OTN transmission, including error counts, alarm information, and signal quality metrics. It helps to isolate issues related to Rx/Tx power loss or OSNR degradation.

Command: display opt trans sfp, Expected Output: SFP/XFP status and optical signal quality information, Description: This command provides detailed information about the SFP/XFP status, including optical signal quality, temperature, and voltage. It helps to identify potential issues related to SFP/XFP degradation or physical damage.
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Resolution Steps

Step 1: Step: Inspect and replace SFP/XFP transceivers, Command: replace opt trans sfp, Description: This step involves inspecting the SFP/XFP transceivers for signs of degradation or physical damage and replacing them if necessary. Ensure that the replacement SFP/XFP transceivers meet the required specifications and are properly seated and secured.

Step 2: Step: Verify fiber connections and splice any damaged fibers, Command: verify opt trans fiber, Description: This step involves verifying the integrity of the fiber connections and splicing any damaged fibers. Ensure that the fiber connections are secure and properly connected to the SFP/XFP transceivers.

Step 3: Step: Adjust Rx/Tx power levels or re-configure OTU4 parameters, Command: adjust opt trans power, Description: This step involves adjusting the Rx/Tx power levels or re-configuring the OTU4 parameters to optimize signal quality and mitigate power loss. Ensure that the adjustments are made within the recommended specifications and do not compromise the integrity of the OTN transmission.

Post-Check Verification

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Command: display opt trans info, Expected Output: OTN transmission information, including optical signal quality and SFP/XFP status, Description: This command provides a comprehensive view of the OTN transmission information, including optical signal quality, SFP/XFP status, and transmission parameters. It helps to verify that the issues have been resolved.

Command: display opt trans error-stat, Expected Output: Error statistics for the OTN transmission, including error counts and alarm information, Description: This command provides detailed error statistics for the OTN transmission, including error counts, alarm information, and signal quality metrics. It helps to verify that the issues have been resolved.

Command: display opt trans sfp, Expected Output: SFP/XFP status and optical signal quality information, Description: This command provides detailed information about the SFP/XFP status, including optical signal quality, temperature, and voltage. It helps to verify that the SFP/XFP transceivers are functioning correctly.
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Roll-back Procedures

- Step: Restore the original SFP/XFP transceivers, Command: restore opt trans sfp, Description: This step involves restoring the original SFP/XFP transceivers if the replacement transceivers did not resolve the issue. Ensure that the original transceivers are properly seated and secured.

- Step: Re-splice the fiber connections, Command: re-splice opt trans fiber, Description: This step involves re-splicing the fiber connections if the original splicing did not resolve the issue. Ensure that the fiber connections are secure and properly connected to the SFP/XFP transceivers.

- Step: Re-adjust the Rx/Tx power levels or re-configure OTU4 parameters, Command: re-adjust opt trans power, Description: This step involves re-adjusting the Rx/Tx power levels or re-configuring the OTU4 parameters to their original settings if the previous adjustments did not resolve the issue. Ensure that the adjustments are made within the recommended specifications and do not compromise the integrity of the OTN transmission.